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Mobile apparatus with computer vision elements for classifying shelf-space

Abstract

Disclosed are systems and techniques for determining out of stock conditions on shelves. The techniques can include receiving, by a computing system, image data from a camera having pixel locations that each uniquely address and store a pixel value, generating a backing map having cell locations that each uniquely address and share a unique address with a corresponding pixel location in the image data, each cell location storing a backing value being an empty value if the pixel value is classified as showing the backing of a shelf and the backing value being a nonempty value if the pixel value is classified as not showing the backing of the shelf, determining, in the backing map, a shelf area representing a location of the captured shelf, and identifying an empty area by finding an area above the shelf area containing a threshold number of cell locations storing the empty value.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Application Ser. No. 63/299,481, filed on Jan. 14, 2022. The disclosure of the prior application is considered part of (and is incorporated by reference in) the disclosure of this application.

TECHNICAL FIELD

(1) This document generally relates to technology for computer vision processing for detecting inventory conditions, such as determining if a shelf in a retail store is empty.

BACKGROUND

(2) Computer vision tasks include operations for acquiring, processing, analyzing and understanding digital images, and extraction of high-dimensional data from the real world in order to produce numerical or symbolic information, e.g. in the forms of decisions. Understanding in this context includes the transformation of visual images (the input of the retina) into descriptions of the world that make sense to processes and can elicit appropriate action. This image understanding can be seen as the disentangling of symbolic information from image data using models constructed with the aid of geometry, physics, statistics, and learning theory. The image data can take many forms, such as video sequences, views from multiple cameras, multi-dimensional data from a 3D scanner, or medical scanning device.

(3) Physical inventory or stock can include the physical goods and/or materials that a business currently has available for use, purchase, or consumption. For example, the physical inventory in a retail store can include the items that are either stocked on the store's shelves available for customers to purchase, or that are available in the store's stock room for restocking in instances of low or out of stock conditions on the shelves. Such physical inventory for a retails store may not include inventory items that have already been purchased by customers, and in some instances,

may also exclude inventory items that have been gathered by customers (e.g., placed in shopping cart) but not yet purchased. Physical inventory can vary in other contexts, though.

SUMMARY

(4) This document generally describes technology for more accurately, efficiently, and unobtrusively determining current stock levels of physical items on shelves with computer vision. For example, determining the current stock/inventory levels for physical items has been a long-standing challenge for entities, such as retail stores. Often it has involved performing manual counts of inventory that are available on shelves, but given the labor expense associated with that technique, such counts may not be possible to be performed frequently (e.g., performed once per day). Other techniques have involved specialized shelf equipment, such as sensors and other equipment capable of electronically detecting current inventory levels on shelves. However, such specialized equipment can be expensive to implement across a larger retail store, may require significant efforts to configure and maintain, and may be prone to malfunction/breaking in the event that customers, employees, vendors, or other relevant users disrupt the equipment. Other techniques have relied inventory tracking server systems that correlate data from manual inventory counts, restocking events, and item sales to determine current inventory levels. However, such server systems can often fail to account for human variation injected into a retail environment, such as when customers pick up an item from its designated shelf and place it somewhere else in the store, when customers pick up an item for purchase but have not yet checked out, and inventory shortage events (i.e., theft of items).

(5) The disclosed technology can provide for more frequent and accurate inventory condition detection, such as out of stock conditions (e.g., no inventory items available in designated shelf location for item), over manual and server-based inventory tracking, and without the added expense and complexity of shelf equipment through the use of a mobile apparatus that can optically detect and identify inventory conditions for items in an environment, such as retail store, warehouse, or other appropriate physical environment. Such a mobile apparatus can be incorporated as part of, affixed to, or otherwise mounted on movable structures/apparatus/devices that are already present and frequently used in such environments, such as shopping carts, order picking carts, restocking carts, cleaning devices (e.g., floor sweeping/cleaning machines), and/or other movable structures/apparatus/devices-permitting for unobtrusive stock condition detection as the preexisting movable structures/apparatus/devices to which the mobile apparatus is part of/affixed/attached to is moved throughout an environment (e.g., moved throughout aisles of store) during any time of day (e.g., during hours while store is open, during hours while store is closed, etc.). Furthermore, such a mobile apparatus can accomplish this through the use of two different computer vision systems that are used in combination to first identify instances of inventory conditions (e.g., out of stock conditions, low inventory conditions, and/or other inventory conditions) and then second to determine specific physical products that correspond to those identified inventory conditions—permitting for accurate inventory condition detection in a manner that is computationally efficient and capable of real time processing by a relatively low power edge computing device (e.g., low processor capacity and memory, such as provided by a Raspberry Pi device).

(6) For example, the disclosed technology can include using machine-learning classifiers to determine an image of a shelf shows the shelf to be empty or not. This process can including classifying the image, on a pixel-by-pixel basis, as either showing the back of the shelf area or not. For example, a store may have shelves for items, and behind the shelf may be a solid wall of a particular color, a pegboard, etc. When an image of the shelf and surrounding area is captured, each pixel can be classified by a classifier trained to identify the wall or object at the back of the shelf. The then, the process can identify areas of the shelf, areas of the items on the shelf, and areas where the back of the shelf is visible. In areas where the back of the shelf is visible, the process can identify this location as an empty shelf. Then, a message can be sent to an inventory management server with the location of the empty shelf. Instructions to restock the shelf can be generated,

advantageously allowing for faster and more reliable inventory instructions.

(7) One or more embodiments described herein can include an apparatus for determining an out of stock condition on a shelf, the apparatus including: a camera in data communication with one or more processors, the one or more processors, a mobile power source providing power to the camera and to the one or more processors, a network interface for sending and receiving messages with physically remote destinations over a data network, and computer memory storing instructions that, when executed by the one or more processors, cause the one or more processors to perform operations. The operations can include receiving image data from the camera that includes pixel locations, each pixel location being uniquely addressed in the image data and storing a pixel value captured by the camera at that location, generating a backing map including cell locations, each cell location being uniquely addressed in the backing map and sharing a unique address with a corresponding pixel location in the image data, each cell location storing a backing value being an empty value if the pixel value is classified as showing the backing of a shelf and the backing value being a nonempty value if the pixel value is classified as not showing the backing of the shelf, determining, in the backing map, a shelf area representing a location of a shelf captured by the camera, and identifying, in the backing map, an empty area by finding an area above the shelf area containing a threshold number of cell locations storing the empty value.

(8) The embodiments described herein can optionally include one or more of the following features. For example, the operations further can include determining a shelf-identifier for a shelf that is empty of items based on the empty area in the backing map. The operations can also include identifying a barcode target area in the shelf area based on the location of the empty area and reading, in the image data, a barcode located in the image data based on the location of the barcode target area in the shelf area. The operations may also include transmitting to an inventory server system through the network interface an empty-shelf message that includes the shelf-identifier. Moreover, the operations can include generating, using the shelf-identifier, instructions to restock the shelf that is empty of items.

(9) In some implementations, generating the backing map can include, for each cell location, accessing the pixel value of the corresponding pixel location with the cell location's unique address, generating a backing value by providing, to a classifier, the pixel value of the corresponding pixel location, receiving, from the classifier, the backing value, and recording, in the cell location, the backing value. Generating a backing value by providing, to a classifier, the pixel value of the corresponding pixel location further can include providing the image data and the unique address of the cell location, and the classifier can be configured to generate the backing value using a model that receives, as input, at least i) the pixel value of the corresponding pixel location and ii) other pixel values in the image data other than the pixel value of the corresponding pixel location based on the unique address. Moreover, generating a backing value by providing, to a classifier, the pixel value of the corresponding pixel location further can include providing a facility location that defines i) where in a facility the image data was captured and ii) item location data that specifies shelves in the facility and objects to be stored on the shelves in the facility, and the classifier can be configured to generate the backing value using a model that receives, as input, at least i) the pixel value of the corresponding pixel location and ii) objects to be stored near the facility location that defines where in a facility the image data was captured. The classifier can be a machine-learning classifier. The apparatus can be in data communication with a server system that can receive records of false-empty results, generate supplemental training data from the records of false-empty results and retrain the classifier using the supplemental training data. Retraining the classifier using the supplemental training data can include retraining, by the server system, the classifier using at least some original training data that was used to train the classifier in generating the false-empty results.

(10) As another example, the apparatus can also include a fixed camera being fixedly mounted on the cart at a first angle, the camera can be a high resolution camera being controllably mounted on

the cart, the high resolution camera can capture high resolution images and engage at least one of pan, tilt, and zoom operations in response to engagement instructions received from a high resolution camera controller, and a high resolution camera controller. The high resolution camera controller can receive high resolution camera instructions and responsive to receiving the high resolution camera instructions, send the engagement instructions to the high resolution camera. The operations can also include receiving, from the fixed camera, first image data that captures a first inventory object, determining, from the first image data, a spatial location of a first inventory object, generating high resolution camera instructions that are configured to cause the high resolution camera to capture the first inventory object, transmitting the high resolution camera instructions to the high resolution camera controller, and receiving, from the high resolution camera, the image data. Moreover, the high resolution camera can operate in multiple modes, the modes including at least i) a sleep mode and ii) a working mode in which the working mode can consume more power than the sleep mode, and the high resolution camera controller can transition, in response to receiving the high resolution camera instructions, the high resolution camera from the sleep mode to the working mode and to later transition the high resolution camera from the working mode to the sleep mode after the image data is received from the high resolution camera.

(11) One or more embodiments described herein can include a method for determining an out of stock condition on a shelf, the method including receiving, by a computing system, image data from a camera having many pixel locations, each pixel location being uniquely addressed in the image data and storing a pixel value captured by the camera at that location, generating, by the computing system, a backing map having many cell locations, each cell location being uniquely addressed in the backing map and sharing a unique address with a corresponding pixel location in the image data, each cell location storing a backing value being an empty value if the pixel value is classified as showing the backing of a shelf and the backing value being a nonempty value if the pixel value is classified as not showing the backing of the shelf, determining, by the computing system and in the backing map, a shelf area representing a location of the shelf captured by the camera, and identifying, by the computing system and in the backing map, an empty area by finding an area above the shelf area containing a threshold number of cell locations storing the empty value.

(12) The embodiments described herein can optionally include one or more of the following features. For example, the method can include determining, by the computing system, a shelf-identifier for a shelf that is empty of items based on the empty area in the backing map. The method can also include identifying, by the computing system, a barcode target area in the shelf area based on the location of the empty area and reading, by the computing system and in the image data, a barcode located in the image data based on the location of the barcode target area in the shelf area. The method may also include transmitting, by the computing system and to an inventory server system, an empty-shelf message that includes the shelf-identifier.

(13) In some implementations generating, by the computing system, the backing map can include, for each cell location, accessing the pixel value of the corresponding pixel location with the cell location's unique address, generating a backing value by providing, to a classifier, the pixel value of the corresponding pixel location, receiving, from the classifier, the backing value, and recording, in the cell location, the backing value. The classifier can be a machine-learning classifier.

(14) One or more embodiments described herein can include a system for determining an out of stock condition on a shelf, the system including one or more processors and computer memory storing instructions that, when executed by the one or more processors, cause the one or more processors to perform operations that include receiving image data having many pixel locations, each pixel location being uniquely addressed in the image data and storing a pixel value captured by a camera at that location, generating a backing map having many cell locations, each cell location being uniquely addressed in the backing map and sharing a unique address with a corresponding pixel location in the image data, each cell location storing a backing value being an

empty value if the pixel value is classified as showing the backing of a shelf and the backing value being a nonempty value if the pixel value is classified as not showing the backing of the shelf, determining, in the backing map, a shelf area representing a location of the shelf captured by the camera, and identifying, in the backing map, an empty area by finding an area above the shelf area containing a threshold number of cell locations storing the empty value.

(15) The systems, devices, program products, and processes described throughout this document can, in some instances, provide one or more of the following advantages. For example, a cart that is being moved through an environment for a primary reason such as storing a shopper's items or scrubbing the floor can be extended to unobtrusively and automatically perform inventory-monitoring tasks. By including cameras and data components, these tasks may be performed without any particular input needed from the user of the cart and can instead passively collect data. This can improve the use of the cart without imposing costs of complexity or attention on the user. By pushing computations related to inventory management to the network edge in the form of a fleet of carts, an inventory system can decentralize the data processing tasks, reducing the load on key, central components. Using high resolution cameras, which can generate images requiring more computational resources to process and may consume more power, only when lower resolution cameras initially identify an item of interest can allow the system to more efficiently use computational resources and to use less battery power than other configurations which may, instead, engage high resolution cameras at all times and for all tasks. This can be particularly beneficial for devices like carts which are usually not tethered to a power source and must instead carry batteries. By using small classifier models, few computing resources may be needed to classify pixels as showing an empty shelf area, further reducing the energy demands of the system and extending battery life.

(16) In another example, the apparatus can be built on top of and integrated with existing in-store processes. Although the apparatus could be incorporated as part of an autonomous standalone robot, it is able to be incorporated into and used with existing devices in the store, such as a human-pushed cart, which can avoid introducing additional devices into stores, like a robot, that may block aisles or otherwise create obstacles for shoppers. Moreover, processing can be performed at the apparatus itself with edge computing, which can avoid clogging network bandwidth, increase efficiency, and use less RAM and processing power. The executable for image processing can also be small in size and more easily deployable at the controller of the apparatus.

Description

DESCRIPTION OF DRAWINGS

- (1) FIG. 1 shows an example system for analyzing images of shelves.
- (2) FIG. 2 shows example data used in analyzing images of shelves.
- (3) FIG. 3 shows an example process for analyzing images of shelves.
- (4) FIG. 4 shows an example process of operating a cart with features to analyze images of shelves.
- (5) FIG. 5 shows an example system of computing devices in a retail location.
- (6) FIG. 6 shows an example system that generates classifier models for image analysis.
- (7) FIG. 7 shows an example process for training classifier models.
- (8) FIG. 8 shows an example system using computer vision to identify stock levels of items on shelves.
- (9) FIG. 9 shows an example of computing hardware of a cart for the collection and communication of stock levels of items on shelves.
- (10) FIGS. 10 and 11 depict an example apparatus affixed to a floor sweeping machine to detect physical inventory conditions.
- (11) FIG. 12 shows an example camera capturing image data of shelves in a retail environment.

- (12) FIG. 13 depicts example data used in analyzing images of the shelves in FIG. 13.
- (13) FIG. 14 is a flowchart of an example process for determining a buffer region in image data of a shelf.
- (14) FIGS. 15A-B show example configurations of an apparatus on a cart to detect inventory conditions.
- (15) FIG. 16 is a flowchart of an example process for determining whether to use image data or RFID sensor data to detect inventory conditions.
- (16) FIG. 17 is a schematic diagram that shows an example of a computing device and a mobile computing device.
- (17) Like reference symbols in the various drawings indicate like elements

DETAILED DESCRIPTION

- (18) As described above, this document describes technology that can identify inventory levels in storage areas like shelves or bins. Low-resolution cameras can capture low-resolution images of a shelf, and a controller can make an initial determination to identify possible areas where a background surface, instead of inventory items, is detected. The controller can instruct a high-resolution camera to capture images of the same area, and those high-resolution images can be used to analyze the area to identify the product with the detected inventory condition (e.g., out of stock level for product).
- (19) In some configurations, a cart can perform image analysis on hardware incorporated in the cart. This analysis can involve examining images of a shelf to determine if the back of the shelf is shown. If so, empty portions of the shelf can be identified by the cart, and a message can be sent from the cart to an inventory-management server alerting the server to the empty portions of the shelf.
- (20) FIG. 1 shows an example system **100** for analyzing images of shelves. In the system **100**, a computer system **102** uses computer vision to determine if a shelf has items on it, or if the shelf is empty. For example, the system **100** may be used in a retail environment such as a store or distribution center to automatically determine if a shelf is empty or not. In other examples, the system **100** may be used in one or more other environments in which inventory management on shelves can benefit from automated analysis (e.g., pharmacies with controlled substances, safety compliance in situations in which items must be returned to their storage location before dangerous operations can be executed, etc.).
- (21) The computer system **102** is connected to a camera **104** that captures images. The camera **104** may be a stationary camera or movable camera collecting monochromatic images or full color images. In the image, a shelf **106** is captured. In some portions of the image, the shelf is empty and the camera has an unconcluded view of backing **108**. The backing may be, for example, a solid sheet, a pegboard, or other appropriate material. In addition, some portions of the image capture an item **110** (e.g., retail merchandise sitting on the shelf) where the shelf is not empty. In this case, since the backing **108** is occluded, the item **110** is captured in the image. As such, the image partially captures an empty shelf area and partly captures a nonempty shelf area. Further examples of systems capable of capturing images are described later.
- (22) The computer system **102** can classify **112** each pixel of the image from the camera **104** as showing the backing **108** or not. For example, the computer system **112** can submit pixels of the image to a classifier that can classify the pixel. This classification may be in the form of a binary determination (e.g., empty value vs nonempty value) may be in the form of a confidence value (e.g., near 0 for likely empty vs near 1 for likely nonempty), or in another format.
- (23) The computer system **102** can find a shelf **114** location in the image. For example, the computer system **102** can examine the pixel classifications to find a long, generally horizontal shape of nonempty pixels and determine that this must be where the shelf **106** is captured in the image. As will be understood, the computer system **102** can identify different shapes if differently shaped shelves **106** are used. For example, a basket or bin may be used and the computer **102** can

identify wider rectangular or trapezoidal shapes.

(24) The computer system **102** can identify shapes of pixels classified as empty to determine where the shelf **106** may be empty. For example, if a large shape (e.g., more than a threshold number of pixels, minimum dimensions) directly above the shelf area is classified as empty, the computer system **102** can classify that area as empty. The computer system **102** can also identify areas of the shelf **106** as not empty. For example, a shelf **110** may be twelve inches below another shelf, and may have items that are nine inches tall. Such a situation would result in some pixels above the shelf classified as nonempty, then above those some pixels classified as empty. In such a case, the computer system **102** can identify that an item is on the shelf and that there is empty space above the item, resulting in a determination that the shelf **106** is not empty.

(25) The computer system **102** can determine empty shelf locations and items that are missing from the empty locations. For example, the computer system **102** can use barcodes printed on the shelf **106**, positioning information for the camera, or other data to determine which shelf **106** in a facility was captured by the image. With this location information, the computer system **102** can determine which item is expected to be on the shelf **106** or assigned to the shelf **106**. Then, the computer system **102** can perform appropriate steps in response to determining that the item is not on the shelf **106**. For example, the computer system **102** can generate instructions to restock the shelf, can generate a report about the empty shelf, can order more inventory to be delivered to the facility, etc.

(26) FIG. 2 shows example data used in analyzing images of shelves. A backing map **200** is rendered in FIG. 2 to show data that can be stored in computer memory (e.g., in computer system **102**, memory **906**) in, for example, binary format.

(27) Images received from cameras can include pixels storing pixel values. For example, in a monochrome image, the pixel value can be a value from **0** to **1**, from **0** to **256**, etc. In a full color image, the pixel value can be in the form of a Red-Blue-Green (RGB) value. Each of these pixels can be uniquely addressed in the image with an [X][Y] address. Similarly, the backing map **200** can include a map of cell locations that is uniquely addressed in the backing map **200** with an [X][Y] address. As such, cells of the backing map **200** can have a one-to-one correspondence with pixel locations in an image.

(28) The backing map **200** can be used to store classifications of pixels in an image by storing classification data in a cell with the same address as the pixel. For example, a pixel at location [10][17] may be classified as showing an empty or nonempty space, and an empty or nonempty value (e.g., 0 or 1) can be stored in the cell [10][17] of the backing map. In the example shown here, cells holding an empty value are shown shaded, and cells holding a nonempty value are shown in white.

(29) With a completed backing map **200** for an image, a computer system can perform analysis on the image using the backing map **200** with or without the image. For example, the computer system can examine shapes of cells to identify objects shown in the corresponding image. In some cases, these shapes are contiguous areas of cells all having the same empty or nonempty value. In some cases, a threshold number or percent of the cells may have different values. This can allow the analysis to be performed even when normal amounts of noise are introduced into the image. For example, the image may capture an item on a shelf, and the item may be similar in color to the backing or may be in a shadowy area. In such cases, some of the cells corresponding to the item in the image may be misclassified as empty when in fact they are not empty.

(30) One or more shelf lines may be determined in the map **200**. The shelf lines **202** can be found in locations having long, generally straight, generally horizontal shapes of cells with nonempty values if the shelves being imaged are similarly long, generally straight, and generally horizontal. The shelf lines **202** represent boundaries in the image where the edges of the shelves are located.

(31) A back shape **204** can be identified in areas where a sufficient group of empty cells are found adjacent to, above, or near a shelf line **202**. Shown here, a bounding box is drawn around the back shape. This bounding box may be stored in the map **200**, in another data structure, or never explicitly calculated. As can be seen in the map **200**, not all empty cells are classified as being part

of a back shape. For example, in areas where nonempty cells are above a shelf line **202**, analysis can determine that there is an item on the shelf, and then empty space above the item.

(32) A barcode target **232** can be identified to find an identifier of the items that the shelf normally holds. The identifier can include, but is not limited to, one dimensional (1D) barcodes, two dimensional (2D) barcodes, SKUs, and other types of visual identifiers. A shape that is below the back shape **204** and between the shelf lines **202** can be identified as a barcode target area **232**. This barcode target area **232** can identify an area in the image to be read to find a barcode value to identify which items are to be stocked or stored in the empty area. As will be understood, identification in the map **202** can produce a group of cell addresses, and the pixels having the same addresses in the image can be examined for the barcode. As such, using a map and image with the same unique addresses for corresponding cell and pixel locations can allow for more simple and efficient analysis than other systems in which different address schemes are used.

(33) Other features of the image can be identified by analysis of the map **200**. For example, an item lines **206** can be identified at the upper border of shapes of nonempty cells. The location of these item lines **206** relative to a shelf line **202** can provide an indication of how many items are stacked on the shelf. For example, the map **200** may be generated in a case where a shelf has one box, then no boxes, then three boxes. As such, this height may be used for inventory management to, for example, order restocking of a shelf that is not empty but is below a threshold number of items. This can advantageously provide for stocking 'just in time' or ahead of actual need, providing for continuously available items or continuously nonempty shelves.

(34) FIG. 3 shows an example process **300** for analyzing images of shelves. For example, the process **300** can be used to examine images of shelves to determine if the shelves are empty, and if they are empty, identify items that should be on the shelf. In the process **300**, a backing map such as the backing map **200** is being used. However, other processes may use different data.

(35) Image data is received **302**. For example, a camera is equipped with a sensor that captures light, a distance sensor is equipped with a sensor that measures distance to a solid object, etc. This sensing data can be compiled into image data with pixel locations, and the image data can be transmitted to a computer or controller for analysis.

(36) A backing map is generated **304**. For example, the backing map **200** can be generated with the same number and arrangement (e.g., row count and column count) of cells as the image data has pixel locations. Analysis of the image can be performed and data that results from the analysis can be stored in the cells.

(37) For example, pixel values (e.g., color or intensity value) that correspond to a cell's location can be accessed, and the pixel value can be supplied to a classifier that classifies the pixel as shown either shelf backing (e.g., shelf backing **108**) or an item in front of the backing. Then, an empty or nonempty value can be stored in the cell of the map **200** with the same address as the pixel location.

(38) The classifier may in some cases be a machine-learning classifier that uses a model generated by training a model on a corpus of test images that have been tagged with empty or nonempty tags. The classifier may in some cases be a non-machine-learning classifier that uses a set of rules and heuristics to classify pixels of the image.

(39) In some cases, the classifier may other data as input in addition to the pixel value. For example, the entire image data can be supplied to the classifier along with an address of a pixel location to be classified. In such cases, the classifier may use a model that considers at least i) the pixel value of the corresponding pixel location, and ii) other pixel values in the image data other than the pixel value of the corresponding pixel location based on the unique address. For example, the model may use image recognition on surrounding image areas to determine context for the pixel at issue—e.g., a dark pixel surrounded by light-colored items may be classified as more likely to be empty. As will be appreciated, the particular decision criteria of many machine-learning models are not completely understood or documented.

(40) In some cases, a location at which the image is captured may be provided, along with item location data that specifies shelves in the facility and objects to be stored on the shelves in the facility. In such cases, the classifier may use a model that considers at least i) the pixel value of the corresponding pixel location, and ii) objects to be stored near the facility location that defines where in a facility the image data was captured. For example, the model may use the location information to determine the colors of items scheduled to be placed on shelves near the area of the image and refine the classification based on the colors in the image. For example, a mid-tone pixel value in an area with brightly colored items may be more likely to be classified than a similar mid-tone pixel value in an area with darkly colored items. As will be appreciated, the particular decision criteria of many machine-learning models are not completely understood or documented.

(41) After submitting the input data to the classifier, the classifier can supply a backing value as output, and the backing value may be recorded in a cell in the data **200** with the same address as the pixel being classified.

(42) Shelf areas are identified **306**. For example, the backing map **200** can be examined to identify shapes of nonempty cells that match shapes of the shelves. In cases where the shelves are thin and flat, thin flat rectangles may be identified as shelf areas. In cases where the shelves include bags or baskets, shapes such as trapezoids and semicircles may be identified as shelf areas.

(43) Empty areas are identified **308**. For example, the backing map **200** can be examined to find areas of empty values that meet one or more rules. The rules may in some cases be generated with machine-learning by training a model on a corpus of test maps that have been tagged with empty or nonempty areas tags. The rules may in some cases be generated with a non-machine-learning set of rules and heuristics to classify pixels of the image. For example, the rules may identify as an empty area groups of pixels directly above a shelf area.

(44) A shelf identifier for the empty area is determined. For example, the image data and/or other data may be used to determine the shelf that is empty, the items to be placed on the shelf, etc. In some cases, this can include identifying a barcode target area within or relative to (e.g., above, below, within a threshold distance) the shelf area, and scanning the image data in the barcode target area to read a barcode captured by the image data.

(45) FIG. 4 shows an example process **400** of operating a cart with features to analyze images of shelves. In the process **400**, a cart is equipped with a fixed camera **402**, a cart controller **404**, and a pan-tilt-zoom (PTZ) controller **406** that controls a PTZ camera. The cart is in data communication with an inventory server system **408**. Further details of these elements can be found later in this document. However, other devices and systems can be used to perform the process **400** and other processes. Moreover, although the process **400** is described in reference to a PTZ camera, the process **400** works with any other type of camera, such as a high resolution camera.

(46) A fixed camera **402** sends **410** image data to the cart controller **404**. The cart controller **404** identifies an inventory item of interest in the data from the camera **402** to collect higher-detail data with the PTZ camera. For example, the fixed camera may be operated continuously while the cart controller **404** is engaged and the cart controller **404** can monitor the data from the fixed camera **402** to identify areas that may have an empty area **308**. When such an area is found, the cart controller can identify that as an area of interest to be more closely examined with a higher resolution, higher power PTZ camera.

(47) To do so, the cart controller **404** can look up **414** an offset value that defines a difference in location between the fixed camera **402** and the PTZ camera. For example, the cart controller **404** may maintain in memory a list of offsets and their associated fixed camera identifier, if there are more than one. The cart controller **404** can use this offset to modify the location in the view space of the fixed camera **402** by, for example, multiplying the location in 3D space by a matrix that defines a translation and transformation.

(48) The cart controller **404** can then generate **418** PTZ instructions using the modified location in order to instruct the PTZ camera to pan, tilt, and zoom to capture the location of the object of

interest. The PTZ controller **406** is configured to receive **418** PTZ instructions from the cart controller **404**. For example, the cart controller **404** can send, over a network of the cart, the PTZ instructions to the PTZ controller **406**.

(49) The PTZ controller **408** is configured to, responsive to receiving the PTZ instructions, send **420** the engagement instructions to the PTZ camera. For example, the PTZ controller **406** can drive the zoom motor in accordance with zoom commands received, can drive the pan moto in accordance with the pan instructions received, and can drive the tilt motor in accordance with the tilt commands received.

(50) In some cases, the PTZ controller **406** can also wake the PTZ camera up from an energy saving state. For example, the camera PTZ camera be configured to operate in a plurality of modes, the plurality of modes comprising at least i) a sleep mode and ii) a working mode, wherein the working mode consumes more power than the sleep mode. As such, the PTZ camera can be kept in the sleep mode to conserve battery life of the cart when not needed, and only selectively be awoken into the working mode when needed. In such cases, the PTZ controller **406** is configured to transition, in response to receiving the PTZ instructions, the PTZ camera from the sleep mode to the working mode and to later transition the PTZ camera from the working mode to the sleep mode after the PTZ image data is transmitted from the PTZ controller to the cart controller.

(51) The PTZ controller **406** is configured to return **422** the PTZ image data and the cart controller **404** is configured to receive **424** PTZ image data. For example, the PTZ controller **406** can send, over the cart's network, images captured by the PTZ camera. These images may be higher resolution, use a larger color space, and/or be captured at a greater frequency than the images from the fixed camera **402**.

(52) The cart controller **404** can analyze the image data to identify **426** empty shelf areas. One example process **300** for doing so is described previously. If an empty shelf area is found **428**, the cart controller **404** can transmit to the inventory server system **408**, through a network interface, an empty-shelf message includes the shelf-identifier **430**. For example, the message may include a barcode value or similar that is associate with the empty shelf location. When information is transmitted to the inventory server system **408**, a bundle of information can be transmitted. The bundle of information can include location information and one or more image data of the empty shelf location. The bundle of information can also include an inference of whether the shelf location is empty (e.g., an out of stock condition). This bundle of information can be used by the inventory server system **408** to validate the determination made by the cart controller **404** that the shelf location is empty. Accordingly, the inventory server system **408** can perform additional processing and analysis of the image data to verify that the shelf location is in fact empty.

(53) The inventory server system **408** can generate, using the shelf-identifier, instructions **432** to restock the shelf that is empty of items. For example, the inventory server system **408** can generate instructions to be rendered on a human user's computing device with a request to pick an item from a back room and to place the item in the empty shelf location. The instructions can include or be computer-readable instructions for one or more automated item-handling devices. For example, the empty shelf may be in a warehouse with automated package handling equipment, and the instructions may instruct a conveyor belt, etc., to move an item to the empty shelf space.

(54) Although FIG. **4** is described in reference to a PTZ camera, the process **400** can also be performed using one or more different types of cameras, such as cameras that are capable of panning only, zooming only, tilting on, adjusting aperture only, adjusting shutter speed only, or any combination thereof.

(55) FIG. **5** shows an example system **500** of computing devices in a retail location. The system **500** may be used, for example, by a store or distribution center that has carts moving around shelves on which items are stored. This can include, for example, shopping carts and floor scrubbers moving around a retail store with products on shelves that are picked up by customers and taken to a cash register for purchase. In another example, a distribution center can have pallet

jacks and carts that are used to stock shelves and fulfill orders for the items on the shelves. In any case, it will be appreciated that items can be placed and picked up off the shelves, and the system **500** can be used to allow for automatic collection of data about the status of those shelves or other storage areas (e.g., bins, cartons, bags, pallets).

(56) Cart controllers **502** and other data-processing hardware on the carts can be used to perform image analysis on the shelves and items in order to identify shelves that are empty. As will be appreciated, these cart controllers **502** may be configured to operate using mobile power sources (e.g., batteries, renewable sources) and may not be tethered to stationary power sources such as electrical mains. Further, a local network **504** connects the cart controllers **502**, an inventory server system **506**, and other devices (e.g., computers and phones). Like all networks, the network **504** has limited bandwidth and capacity. As such, the system **500** can be advantageously configured so that the cart controllers **502** transmit only small messages **508**. The small messages **508** may be configured to include only alpha-numeric data and not larger data like images collected by cameras on the carts. For example, the cart controllers **502** can perform image analysis process (e.g., process **300** and **400**) on the network edge, transmitting only the results in the small messages **508** and not the full images or maps. Compared to the full images or maps, the small messages may specify, for example, a shelf found to be empty and a timestamp. This may allow for the use of small messages **508** that are orders of magnitude smaller than the images that were analyzed to produce the small messages **508**. As will be understood, a system with many components transmitting small messages instead of large messages is a useful advantage to the computing technology. Similarly, cart controllers **502** using small classifier models for image classification tasks can contribute to these advantages by requiring less power from the mobile power sources, which, like all mobile power sources, are limited in the amount of power they can provide before running out, needing recharging, etc. Systems and process for generating such small models will be described later.

(57) FIG. **6** shows an example system **600** that generates classifier models for image analysis. For example, the process **600** can be used to generate classifier models for use in the processes and system described in this document to classify pixels as showing an empty or nonempty shelf space, to identify shapes on a backing map, and for other purposes.

(58) One or more training servers **602** can perform machine learning processes on training data to create models that classify later-received data in a way that the training data classifies training data. For example, a training controllers **604** can be used to collect large image files **606** of shelves that empty as well as shelves that are nonempty in real or simulated environments (e.g., real retail stores or areas in an engineering lab with shelves and products placed as they would be in real retail stores). The training controllers **604** may be incorporated into carts as described in this document, or may be used in other (e.g., lower cost) rigs. Training data for the large image files **606** can also be provided to the training servers **602**. The tagging data **608** can tag each large in file **606**, or subsections of the images, as having an empty shelf or a nonempty shelf. In some cases, the tagging data **608** can be generated by human users reviewing each image and recording data in a computer to tag the image. In some cases, the tagging data may be generated in whole or in part based on existing corpuses of images that have already been tagged by a different process, for another purpose, etc.

(59) The training server **602** can generate a small classifier model **610** for distribution to controllers in use **612**. For example, the training server **602** can communicate the small classifier model to cart controllers already in retail environments to update old classifier models. In some cases, the small classifier model **610** can be incorporated onto a disk image used to manufacture new cart controllers and/or new carts. The training server **602** may transfer packets of data instead of entire streams of data (e.g., image files). The training controller **604** can operate like a filtering mechanism to filter out what data is transmitted between components, such as what information is distributed to the controllers in use **612**. Transferring packets of data can be beneficial because it leads to more efficient data transmission and a reduction or avoidance in clogging network

bandwidth. The controllers can then process the data in the packets both faster and more efficiently when utilizing local hardware and computational resources.

(60) The controllers in use **612** can use the small classifier model **610** while performing image analysis tasks, such as those described in the processes **300** and **400**. In some cases, the use of this technology can lead to incorrect results. For example, a cart controller **502** may report to the inventory server system **506** a message **508** that a particular item is out of stock on a shelf. However, when a worker is dispatched to restock that item, they may discover that the item is already on the shelf. Such false-empty events can be logged in a false-empty report **614** and communicated to the training server **602**.

(61) The training server **602** can use one or more false-empty reports to retrain the model and produce a new small classifier model **610** that is more accurate and can account for the conditions that created the false-empty report. This updated model **610** may then be distributed to the controllers in use **612** and the controllers in use can begin using the updated model **610** instead of the older model **610**. This can provide for classifier models that advantageously improve over previous versions, allowing for more accurate image processing and allowing for updated models in changing conditions.

(62) As shown, the small classifier model is distributed to controllers **612** for use in image analysis processes. In some configurations, the small classifier model can be recorded in computer memory or transmitted over data networks using binary data of a size on the order of 500 kilobytes (kB), less than 1 megabyte (MB), etc. One advantage of using models of this small size is that distributing the model to many destinations uses fewer network resources than other technology that must transmit much larger model. This advantage is compounded when the small classifier model **610** is transmitted to many destinations as would be the case for a retail enterprise with hundreds or thousands of facilities, each with dozens or hundreds of cart controllers.

(63) Another advantage to the use of small classifier models **610** is that they may be used by cart controllers that have limited processing power and energy availability. This can allow image processing to be performed at network edge locations (cart controllers) instead of in central computing resources like an inventory management server. Performing the image processing at the edge allows for capacity to expand the system, as each cart controller added to the system brings along the computational resources required to facilitate the additional image processing required for the cart controller. A small model also allows for processing to be performed on hardware of limited computational power. Such controllers provide a number of advantages including lower cost than more powerful hardware, and lower power consumption. As explained in this document, the controllers can be powered by a mobile power sources such as a battery which can have a limited capacity. By using lower models, and thus less battery power, cart controllers using this technology can perform more image recognition tasks between charging sessions.

(64) FIG. 7 shows an example process **700** for training classifier models. For example, the process **700** can be performed by the system **600** and thus system **600** will be used in describing the process **700**. However, another system or systems can be used to perform the process **700** or similar processes.

(65) Records of false-empty results are received **702**. For example, the controllers **612** can report an item as not on a shelf, and an inventory server system can determine that the item was in fact on the shelf when the cart controller analyzed that shelf. In such cases, the inventory server system can generate a false-empty record that includes a copy of the image that was analyzed, requested from the cart controller **612** after the false-empty result was identified, and other data such as timestamp, facility location, etc.

(66) Supplemental training data is generated from the records of false-empty results **704**. For example, using human tagging data and/or data from the false-empty result, the training server **602** can generate supplemental training data that is in the same format as the large image files **606** and tagged data from training **608** previously used to train the small classifier model.

(67) Classifier models are retrained using the supplemental training data **706**. For example, the training server can use all of the supplemental training data along with some, all, or none of the original training data **606** and **608** that was used to train the classifier that was used in generating the false-empty results. In some cases, use of all or some of the original training data along with the supplemental training data can improve the retraining process, and in some cases, using exclusively the supplemental training data may instead improve the retraining process.

(68) Retrained classifier models are distributed **708**. For example, the training server **602** can transmit the new small classifier model **610** to the controllers **612** over data networks, and new controllers being created can be programmed initially with the update small classifier model.

(69) FIG. **8** shows an example system **800** using computer vision to identify stock levels of items **820** on shelves **802**. In the system **800**, a cart **806** (example of a movable device that apparatus can be a part of or otherwise attached to) is traveling through an aisle or other area of an environment (e.g., retail store) with shelves **802**. A fixed camera **810** on the cart **806** can generate low-resolution images of the shelves **802**. A movable high-resolution camera **812**, such as a Pan Tilt Zoom (PTZ) camera, can also be included on the cart **806** to generate high-resolution images of specific locations on the shelves **802**.

(70) The shelves **802** can store inventory **822** for sale, storage, staging, etc. The shelves **802** can include a backer surface **814** that forms a vertical wall defining a back of each of the shelves **802**. When items **820** are on the shelves **802**, the items **820** can occlude view of portions of the backs **814**, and when items **820** are absent from the shelves **802**, such as at location **824** and location **826** on the shelves **802**, more of the backs **814** can be viewed from the perspective of the cart **806**. The shelves **802** can additionally include labels **822** that identify the items **820** and that are positioned adjacent to the items **820** on the shelves **802** (e.g., affixed to front surface of shelf). The labels **822** can include, for example, a product description (e.g., product name, manufacturer name, model number), a product price, and one or more unique identifiers for the product, such as a barcode or QR code that provides a UPC code for the product.

(71) The cart **806**, shown as **806A** earlier and **806B** later as the cart moves forward, travels past shelves **802**. The cart **806** can be any sort of cart or other device that can be moved through a retail or inventory environment. Examples include, but are not limited to, shopping carts, pallet jacks, floor cleaners, lifts, autonomous inventory-moving robots, etc. In many cases, the cart **806** can include wheels, a handle or other fixture for moving the cart, and hardware (e.g., baskets, motors, scrub brushes, bags, forklift tines) for purposes other than facilitating imaging of the environment (e.g., transporting inventory, cleaning floors).

(72) The fixed camera **810** can be non-movably affixed to the cart **806** to capture images of the environment around the cart **806**. This non-movable affixing can include, for example, the camera **810** being encased in integral housings of the cart, being fastened with fasteners (e.g., screws, nuts-and-bolts). The fixed camera **810** can be mounted on the cart **806** at any of a variety of different orientations, such as angled to the side, forward, backward, and/or other orientations. For example, the camera **810** may be mounted orthogonal (i.e. at a right angle to) the direction of travel to the cart. As will be understood, other angles are possible, including angles greater than orthogonal.

(73) The movable camera **812** (e.g., PTZ camera, a high resolution camera) is mounted to the cart **806** in a cradle with controllable mechanisms (e.g., motors) capable panning and tilting the camera, and the movable camera **812** can further be equipped with controllable structures to optically zoom (e.g., lenses and/or mirrors). As shown, the movable camera **812** can be mounted in a position in the cart **806** such that the movable camera **812** has a point of view into the viewspace of the fixed camera **810** mounted on the cart **806**. For example, the movable camera **812** can be mounted adjacent or near the fixed camera **810**, such as on the same side of the cart **806** as the fixed camera **810** and within a threshold distance of the fixed camera **810** (e.g., within threshold horizontal distance, within threshold vertical distance, within overall threshold distance). In some cases, the movable camera **812** may be mounted in a common vertical plane or a common horizontal plane as

the fixed camera **810**.

(74) A controller **808** can also be provided that is communicatively coupled (e.g., wired connection, wireless connection) to both the fixed camera **810** and the movable camera **812**. For example, the controller **808** can receive first images **840** from the fixed camera **810** and second images **842** from the movable camera **812**. The first images **840** can be low resolution images that provide a broader view of the shelf **802** and the inventory items **820**, as indicated by the wider angle defined by the dashed lines emanating from the fixed camera **810**. The second images **842** can be high resolution images of more specific areas of the shelf **802** (as indicated by the smaller angle defined by the dotted lines emanating from the movable camera **812**) that are achieved by adjusting the movable camera **812** according to control signals **844** provided from the controller **808** to the movable camera **812**. For example, the controller **808** can transmit control signals **844** that include, for example, signals to adjust pan, tilt, and zoom for the movable camera **812**.

(75) The controller **808** can identify specific items from the shelves **802** that have inventory conditions, such as being out of stock, having low inventory, having at least a threshold level of inventory, and/or other conditions. To accomplish this, the controller **808** can receive the first images **840** from the fixed camera **810**, as indicated by step A (**850**). The controller **808** can analyze the first images **840** to identify stock conditions in the shelf **802**, as indicated by step B (**852**). Such analysis can include, for example, identifying portions of the shelf **802** where at least a threshold area of the back **814** of the shelf **802** is identifiable in the first image from the vantage point of the fixed camera **810**, which can indicate an out of stock condition. For instance, in the depicted example, the controller **808** can detect location **824** (for the cart at time **806A**) and detect location **826** (for the cart at time **806B**) from first images **840** including at least a threshold area depicting the back **814** of the shelf **802**—indicating an out of stock condition at locations **824** and **826**. Other analysis and conditions may additionally and/or alternatively be detected.

(76) Once an out of stock condition is detected, the controller **808** can determine a physical location on the shelf **802** for the stock condition relative to the position of the cart **806** and the cameras **810/812**, as indicated by step C (**854**). For example, the controller **808** can correlate coordinates from the first image **840** (e.g., x, y coordinates) where the stock condition is detected to a physical location relative to the position of the cart **806**. Using the physical location for the stock condition, the controller **808** can generate and transmit the control signals **844** to the movable camera **812**, as indicated by step D (**856**). The control signals **844** can be generated to move and/or otherwise adjust the movable camera **812** so that it is focused in on the area where the stock condition was detected, such as through adjusting one or more motors or other components in the movable camera **812**. For example, the control signals **844** can cause the movable camera **812** to move and adjust so it is oriented on the area around the locations **824** and **826**.

(77) The controller **808** can then receive second images **842** from the movable camera **812** from the area around the target locations (e.g., locations **824** and **826**), as indicated by step E (**858**), and analyze those second images **842** to identify a product that corresponds to the stock condition, as indicated by step F (**860**). Identifying the product corresponding to the stock conditions can include the controller **808** analyzing the second images **842** to detect, for example, identifying information (e.g., barcode, UPC number, product name) from product labels **822** that correspond to locations where the stock conditions are detected. For example, the controller **808** can identify the label **828** that corresponds to the location **824** and the label **830** that corresponds to the location **826**, and analyze portions of the second images **842** depicting those labels **828** and **830** (e.g., image-based barcode recognition, optical character recognition, object detection techniques) to identify the corresponding product. The identifying information can include, for example, a UPC code or other unique identifier for the product.

(78) The controller **808** can include a network interface that is capable of transmitting detected stock conditions along with product identifiers to an inventory server system **870**, which can maintain a database **872** of current inventory levels, as indicated by step G (**862**). The inventory

server system **870** be local and/or remote from the environment (e.g., retail store) where the cart **806** and the shelves **802** are located. For example, the inventory server system **870** and its inventory data **872** may be part of a broader system that is used within an individual retail store by workers to determine when to restock various items **820** on the shelves **802**, and/or may be used by remote customers (e.g., mobile device user, desktop user) looking up current availability of items **820** on the shelves **802** through a mobile app or web interface. As a result, the use of the cart **806** to passively and automatically identify inventory conditions for the items **820** on the shelves **802** can provide more accurate and up-to-date inventory information for both internal users and external users. Furthermore, by adding the controller **808** and cameras **810/812** to a fleet of carts **806** and/or other movable devices/apparatus in an environment (e.g., retail store), most if not all of the environment may be traversed within a period of time (e.g., half hour, one hour, 4 hours, 8 hours) across the fleet of carts **806** and/or other devices/apparatus to provide comprehensive and updated stock condition information for the environment.

(79) The cameras **810/812** and the controller **808** may be part of the cart **806** and/or may be attachable to the cart **806**. In some instances, the cameras **810/812** and the controller **808** can be contained within a common housing that can be mounted to the cart **806**. Other configurations are also possible.

(80) FIG. **9** shows an example of computing hardware **900** of a cart for the collection and communication of stock levels of items on shelves. For example, the hardware **900** may be integrated into a cart **806** or other device.

(81) The system **900** can include a cart controller **902** with one or more processors **904** and memory **906**. The cart controller can be integrated into the body of the cart, including by having an integrated housing made of a rigid material (e.g., plastic, metal). A data network of the cart **908** can pass through the housing, for example with wires passing through one or more ports, and/or with wireless data signals passing through the rigid material.

(82) The data network **908** can include one or more wired and wireless data links that connect components of the system **900** to allow data communication, electrical power transmission, etc. Data network **908** can sometimes be referred to as a control area network (CAN) or a data bus. A network interface **910** can interface with one or more external data networks and allow elements of the system **900** to communicate with elements of other data processing systems including, but not limited to, remote servers, local computing devices such as smartphones, etc. The network interface can include some, all, or none of the following types of network interfaces: BLUETOOTH, WiFi, Zigbee, and Ethernet. The data network **908** can collect components including but not limited to low resolution cameras **912**, a high resolution camera **914**, a high resolution camera controller **916**, zoom motor **922**, pan motor **924**, tilt motor **926**, focus mechanism **928**, shutter speed adjuster **930**, aperture adjuster **932**, focal length adjuster **934**, inertia measurement unit **936**, indoor location device **942**, depth sensor **944**, RFID sensor **946**, light source **948**, output indicator **950**, and/or power source **952**. The components **922**, **924**, **926**, **928**, **930**, **932**, **934**, **936**, **942**, **944**, **946**, **948**, **950**, and **952** can be optional. In some implementations, fewer than all of the components **922**, **924**, **926**, **928**, **930**, **932**, **934**, **936**, **942**, **944**, **946**, **948**, **950**, and **952** may be included.

(83) Low resolution cameras **912** and a high resolution camera **914** can collect image data from the surrounding environment and transmit that image data to the controller **902**. The low resolution cameras **912** can include hardware, firmware, and software capable of capturing images that produce less data (e.g., lower resolutions, at a lower framerate, and/or with fewer color) than the high resolution camera. For example, the low resolution cameras **912** may capture images every 0.5 seconds in monochrome at a resolution of 1224×600, while the high resolution camera **914** can capture images at 60 Hz in 16 bit at 3840×1600. As will be understood, the cameras **912** and/or **914** may be selectively engaged or disengaged. For example, the controller **902** may collect image data from the low resolution cameras **912** at all times, and may turn on or awaken the high resolution camera **914** only selectively. For example, the high resolution camera **914** may be engaged

periodically, in response to the controller **902** identifying an item of interest in data from the low resolution cameras **912**, etc. In such a way, the system **900** can gain the benefits of the use of the high resolution camera **914**, without requiring the electricity to operate the high resolution camera **914** at all times.

(84) The high resolution camera controller **916**, which can include one or more processors **918** and memory **920**) can control one or more operations of the high resolution camera **914**. While not shown here for example, the controller **916** can receive instructions from the cart controller **902** to adjust pan, tilt, and zoom of the high resolution camera **914**. In response, the controller **916** can engage one or more motors accordingly. The zoom motor **922** can engage to zoom the high resolution camera **914** in and out. The pan motor **924** can engage to pan the high resolution camera **914**. The tilt motor **926** can engage to tilt the high resolution camera. As other examples, the focus mechanism **928** can be mechanical and/or digital. The focus mechanism **928** can be engaged by the controller **916** to adjust a focus of the high resolution camera **914**.

(85) The shutter speed adjuster **930** can be mechanical and/or digital. The shutter speed adjuster **930** can be engaged by the controller **916** to adjust a shutter speed of the high resolution camera **914**. Similarly, the aperture adjuster **932** can be mechanical and/or digital. The aperture adjuster **932** can be engaged by the controller **916** to adjust an aperture of the high resolution camera **914**. The focal length adjuster **934** can also be mechanical and/or digital. The focal length adjuster **934** can be engaged by the controller **916** to adjust a focal length of the high resolution camera **914**.

(86) The inertia measurement unit **936** can be configured to determine how much a mobile apparatus (e.g., a cart) moves throughout an environment, such as a retail environment (e.g., a store), what movements are made by the mobile apparatus, where the mobile apparatus has moved, orientation of the mobile apparatus relative to shelves and other structures or locations in the environment, and depth of field or distance between a shelf and the mobile apparatus. Accordingly, the inertia measurement unit **936** can determine orientation, speed, direction of gravity, and direction of movement relative to the direction of gravity for the mobile apparatus. The inertia measurement unit **936** can include a gyroscope **938** and an accelerometer **940** to detect movement of the mobile apparatus and make the abovementioned determinations.

(87) The indoor location device **942** can be an upward facing camera that determines a location of the mobile apparatus in the environment based on what lights are detected in the ceiling of the environment by the indoor location device **942**. For example, the indoor location device **942** can detect lights that are emitted from lightbulbs in the ceiling of the environment and can identify those lights in a lookup table that associates the lights with locations in the environment. The indoor location device **942** can then triangulate a location of the mobile apparatus based on correlating the locations for the lights that are identified via the lookup table. Sometimes, the mobile apparatus may not have the indoor location device **942** and can instead employ a different location positioning system to determine a current location of the mobile apparatus in the environment. For example, the mobile apparatus can utilize Bluetooth beacons or other global positioning devices and signals.

(88) The depth sensor **944** can detect signals or other data that can be used by the controller **916** or the cart controller **902** to determine a depth or distance between the mobile apparatus and a particular shelf, barcode, or other region of interest (ROI) in the environment. The depth sensor **944** can, for example, be a stereoscopic camera. The depth sensor **944** can also use LiDAR and/or infrared sensors to detect depth or distance between the mobile apparatus and a particular shelf, barcode, or ROI.

(89) The RFID sensor **946** can be used to detect and identify inventory conditions for one or more products. When locations in the retail environment are used for storing soft goods or other products that do not have definitive physical structures (e.g., clothes, towels, curtains, etc.), RFID sensor **946** readings can be used to determine whether the soft goods are out of stock. After all, image data may not be as effective in identifying out of stock conditions for products that do not have

definitive physical structures that can be identified in image data. As an illustrative example, the indoor location device **942** or another indoor location system can detect current location of the mobile apparatus. When the device **942** detects that the mobile apparatus is located in a region where soft goods are typically stocked, the indoor location device **942** can send a notification to one of the controllers **902** and **916** to engage the RFID sensor **946**. When the RFID sensor **946** is engaged, the RFID sensor **946** can capture RFID signal data at the mobile apparatus's current location. Such RFID signal data can be used by the cart controller **902** to determine whether there is an out of stock condition at the current location, instead of using image data captured by the high resolution camera **914**.

(90) The light source **948** can be selectively actuated and controlled by the cart controller **902**. For example, the cart controller **902** can analyze one or more images captured by the low resolution cameras **912** and/or the high resolution camera **914** to determine whether the environment is too dark. If the images are dark, grainy, and/or blurry, the cart controller **902** can determine that the light source **948** should be actuated to illuminate an area that is imaged by the high resolution camera **914**. The light source **948** can be an LED light. The light source **948** can also be an infrared light, a red light, or one or more other types of lights that the high resolution camera **914** is sensitive to. Actuating the light source **948** can be advantageous when the high resolution camera **414** is moving at a speed that causes the captured images to appear blurry and/or grainy.

(91) The output indicator **950** can be configured to output an indication of whether components described herein are operating properly. The output indicator **950** can receive indications from one or more of the components described herein that indicates whether operations are being properly executed. The output indicator **950** can then output, for example, differently colored lights to demonstrate to a user of the mobile apparatus (e.g., a store employee) that the components of the mobile apparatus are functioning properly. For example, the output indicator **950** can output a green light to indicate that the components are operating correctly. The output indicator **950** can output a red light to indicate that one or more components are not working properly, thereby informing the user of the mobile apparatus that they should check the components of the mobile device to resolve the issue.

(92) The power source **952** can be a battery (e.g., rechargeable, replaceable, removable) or other source that powers the components of the mobile apparatus. The power source **952** can be part of or integrated into the mobile apparatus. The power source **952** can also be attached to and in communication with components of the mobile apparatus described above.

(93) FIGS. **10** and **11** depict an example apparatus **1000** affixed to a floor sweeping machine **1002** to detect physical inventory conditions. The example apparatus **1000** can be similar to the apparatus discussed above with regard to FIG. **8**. For example, the apparatus **1000** can include a local controller and network device **1004** (similar to the controller **108**), a movable camera **1006** (similar to the movable camera **112**), and a stationary camera **1008** (similar to the fixed camera **110**).

(94) The apparatus **1000** is also depicted as including a location tracker **1010**, which can be communicatively coupled (e.g. wired connection, wireless connection) to the controller **1004** to provide information identifying a current location of the apparatus **1000**, such as the location of the apparatus **1000** within an interior space, global positioning coordinates (e.g., GPS coordinates), and/or other location information. The controller **1004** can be configured to use the location information for any of a variety of purposes, such as using it in combination with images from the cameras **1006** and/or **1008** to detect stock conditions for products on shelves **1012**, and/or using it to report the location where the stock conditions are detected (e.g., reporting stock condition for product with location of apparatus **1000** when condition detected).

(95) The apparatus **1000** is depicted as being attached to a floor sweeping machine **1002** that is manually operated by an operator **1020**. The apparatus **1000** can automatically detect and report stock conditions in the shelves **1012** without any input or direction from the operator **1020**.

Additionally, the apparatus **1000** can detect and report stock conditions in the shelves **1012** passively and without altering the normal operation of the floor sweeping machine **1002**.

(96) The apparatus **1000** is depicted with the movable camera **1006** and the stationary camera **1008** being positioned along a common vertical plane, with the stationary camera **1008** being positioned above the movable camera **1006**. Other arrangements are also possible, such as the movable camera **1006** being positioned above the stationary camera **1008**, the movable camera **1006** and the stationary camera **1008** being positioned side-by-side along a common horizontal plane, and/or other configurations. The stationary camera **1008** may be positioned at a height and orientation that permits for it to capture an image of the full height of the shelves **1012** when the machine **1002** is at one or more target distances from the shelves **1012**. For example, referring to FIG. **11**, the stationary camera **1008** can be positioned so as to have a field of view **1100** that captures the full height of the shelves **1012**.

(97) Still referring to FIG. **11**, the movable camera **1006** can have one or more axes of rotation that can provide the example ranges of motion **1102** and **1104**. For example, the movable camera **1006** can perform a pan motion **1102** permitting the movable camera **1006** to capture images forward or backward along the shelf **1012** relative to the current location of the machine **1002**. The movable camera **1006** can also perform a tilt motion **1104** that can permit capturing images of different vertical positions of the shelf **1012**.

(98) An example out of stock condition is depicted at location **1110**, which the controller **1004** can detect using the images from the stationary camera **1008**. The location **1110** can have a corresponding label **1112** that the controller **1004** can identify and analyze using images from the movable camera **1006** to identify the product that is out of stock. The label can include any of a variety of information that identifies an intended product for location **1110**, including a product name and manufacturer **1114**, a product price **1116**, numerical product identifiers **1118** (e.g., UPC code, retail store identifier, manufacturer identifier), and/or a barcode **1120** that identifies the product (e.g., barcode **1120** encodes numerical product identifiers in optically readable set of symbols).

(99) FIG. **12** shows an example camera **104** capturing image data of shelves **1202**, **1204**, and **1206** in a retail environment **1200**. The camera **104** can be configured to a cart or other mobile apparatus, as described above. As the camera **104** is moved down an aisle in the retail environment **1200**, the camera **104** can be trained on the shelves **1202**, **1204**, and **1206** and configured to capture images of the shelves. These images can then be used (e.g., by the computer system **102**) to determine out of stock conditions of the shelves **1202**, **1204**, and **1206**.

(100) Gaps may exist between bottoms of shelves and backs of shelves because the camera **104** is not always going to have a straight on view of every shelf. Depending on an angle of the camera **104** relative to a shelf, size of the gaps can vary. For example, if the camera **104** is straight on with a shelf (e.g., angled at or close to 180 degrees relative to the shelf), then smaller the gap between the bottom of the shelf and the top of the shelf in image data captured of the shelf by the camera **104**. Moreover, the farther away the camera **104** is from the shelf, the smaller the gap. On the other hand, the larger the angle of the camera **104** relative to a shelf and the farther away from that shelf, the larger the gap between the bottom of the shelf and the top of the shelf in image data of the shelf.

(101) The gap can be defined as a buffer region, which can be ignored when processing the image data so as to more accurately approximate where a back of a shelf actually starts in the image data. Therefore, the computer system **102** can determine (e.g., approximate) the buffer region based on the camera **104**'s angle relative to the shelf and distance away from the shelf. The computer system **102** can then ignore/isolate the buffer region when analyzing the image data to determine an out of stock condition at the particular shelf. This can be beneficial to reduce false positives that may occur if the computer system **102** misidentifies a back of a shelf to include more pixels in the image data than it actually does. Instead, using the techniques described herein, anything beyond the buffer region can be analyzed and processed by the computer system **102** to determine whether

enough of the actual back of the shelf is visible to warrant a determination that the shelf has an out of stock condition.

(102) The buffer region can be identified based on a distance **D** of the camera **104** from the shelves **1202**, **1204**, and **1206**, and an angle of the camera **104** relative to the shelves **1202**, **1204**, and **1206**. As shown in FIG. **12**, the camera **104** has a field of view **1214**, which includes all of the shelves **1202**, **1204**, and **1206**. The camera **104** is positioned on a horizontal plane instead of being tilted up or down at one or more different angles. In some implementations, the camera **104** can be positioned at different angles. In some implementations, the camera **104** can also move between one or more different angles (e.g., the camera **104** can pan or tilt).

(103) Here, the camera **104** is almost straight on (e.g., horizontal plane, 180 degrees) with the shelf **1202**. As a result, a gap **1209** (e.g., buffer region) between the bottom of the shelf **1202** and a back of the shelf **1202** can be very small in image data that the camera **104** captures of the shelf **1202**. When the buffer region **1209** is isolated, the image data of the shelf **1202** can include a larger region **1208** of the back of the shelf **1202**.

(104) The camera **104**'s field of view **1214** is slightly angled from the shelf **1204** in comparison to the shelf **1202**. As a result, less of a back of the shelf **1204** may be visible from the camera **104**'s angle and distance away from the shelf **1204**. Thus, a buffer region **1210** associated with the shelf **1204** can be larger than the buffer region **1209** of the shelf **1202** (e.g., refer to FIG. **13**).

(105) The camera **104**'s field of view is even more angled from the shelf **1206** in comparison to the shelves **1202** and **1204**. As a result, even less of a back of the shelf **1206** is visible from the camera **104**'s angle and distance away from the shelf **1204**. Thus, a buffer region **1212** associated with the shelf **1206** can be larger in the image data than the buffer region **1210** of the shelf **1204** and the buffer region **1209** of the shelf **1202**.

(106) FIG. **13** depicts example data used in analyzing images of the shelves in FIG. **12**. Image data **1300** depicts the shelves **1202**, **1204**, and **1206** as imaged by the camera **104** when the camera **104** is a first distance away from the shelves. When the camera **104** is closer to (e.g., less distance away from) the shelves **1202**, **1204**, and **1206**, the buffer regions **1209**, **1210**, and **1212** can appear larger, as shown in the image data **1300**. The buffer regions **1209**, **1210**, and **1212** can comprise more pixels in the image data **1300**.

(107) On the other hand, when the camera **104** of FIG. **12** is at a second distance away from the shelves **1202**, **1204**, and **1206**, where the second distance is less than the first distance, the buffer regions **1209**, **1210**, and **1212** can appear smaller, as shown in image data **1302**. The buffer regions **1209**, **1210**, and **1212** in the image data **1302** therefore comprise fewer pixels than in the image data **1400** taken by the camera **104** from the first distance.

(108) Furthermore, as shown in FIG. **13**, back of shelf regions **1208**, **1306**, and **1308** appear larger in the image data **1300** than in the image data **1302** because of the distance of the camera **104** from the shelves **1202**, **1204**, and **1206**.

(109) In the image data **1300** taken from the first distance, which is closer to the shelves than the second distance, the back of the shelf **1208** of the shelf **1202** appears larger in pixel size relative to the buffer region **1209** of the shelf **1202**. This is attributed to the angle of the camera **104** relative to the shelf **1202** and the first distance. The back of shelf **1306** of the shelf **1204** also appears larger than the buffer region **1210**, although the buffer region **1210** is larger than the buffer region **1209** of the shelf **1202**. The shelf **1204** can also have a taller height than the shelf **1202**. Finally, the back of the shelf **1308** of the shelf **1206** appears smaller than the buffer region **1212**, which is attributed to angle of the camera **104** relative to the shelf **1206**. The shelf **1206** can also have a smaller height than the shelf **1204** but may be a same or similar height as the shelf **1202**. The image data **1302** is scaled down in size from the image data **1300** since the image data **1302** is captured by the camera **104** from the second distance, which is greater than (e.g., farther away from the shelves) the first distance.

(110) FIG. **14** is a flowchart of an example process **1400** for determining a buffer region in image

data of a shelf. The process **1400** can be performed by the computer system **102**, however, another system or systems can be used to perform the process **1400** or similar processes. For illustrative purposes, the process **1400** is described from a perspective of a computer system.

(111) Referring to the process **1400**, the computer system receives location information in **1402**. The location information can be received from a camera (e.g., the camera **104**) and/or a component of the mobile apparatus or cart that detects movement of the mobile apparatus or cart. For example, the component can be an inertia measurement unit having an accelerometer and/or gyroscope, as described below. The component can also be an indoor locating system or other tracking system that is used in a physical environment where the mobile apparatus or cart is located. The location information can indicate a region in the physical environment where the mobile apparatus (and camera) is located. The location information can also indicate a speed at which the mobile apparatus is moving, images captured by low resolution cameras on the mobile apparatus, where the images can be used to determine distance and/or positioning of the mobile apparatus relative to the shelf. The location information can also include GPS coordinates or other positioning information indicating movement and/or location of the mobile apparatus relative to the shelf.

(112) In **1404**, the computer system determines a distance away from a shelf. The computer system can use the location information to determine or approximate the distance between the mobile apparatus (and the camera) and the shelf. As an example, if the location information includes image data, the computer system can process the image data using one or more machine learning models to measure an approximate distance of the mobile apparatus from the shelf.

(113) For example, the computer system can determine distance away using multiple images (e.g., 2 images). The computer system can receive images from stereoscopic cameras. The stereoscopic cameras can be positioned side-by-side on the mobile apparatus. The computer system can compare the stereoscopic images to determine how aligned imaged features appear in the images. For example, where the features appear further apart or less closely aligned, the computer system determines that the mobile apparatus is closer to the shelf than when the same products appear more closely aligned in the images.

(114) As another example, the computer system can determine distance away based on motion information. The computer system can receive indoor positioning information from components such as an inertia measurement unit that is part of the mobile apparatus. The indoor positioning information can include motion and speed information of the mobile apparatus. Since the camera is maintained at a consistent height, the computer system can determine distance away based on motion of the camera and speed at which the camera (on the mobile apparatus) is moving. Using this motion information, the computer system can compare the images captured at different times and determine the distance away.

(115) As yet another example, the computer system can determine distance away and depth using disparity maps.

(116) The computer system then determines an angle of the camera relative to the shelf (**1406**). The angle of the camera can already be known, as the camera can be calibrated in a fixed position. Since the camera is in a fixed position, the computer system can look at a vertical field of view of the camera, depth in images captured by the camera, and pixels appearing in those images to determine the angle of the camera. The computer system can also determine the angle using the location information. For example, if the location information includes image data, the computer system can process the image data using one or more machine learning models or other programming techniques to measure an approximate angle at which the image data is captured.

(117) In **1408**, the computer system determines a buffer region based on the distance and angle determinations. The computer system can, for example, determine an area of the image data that is expected to be empty. The computer system can estimate a size of a lower portion of the image data (which may represent the empty area), by determining a distance away from the shelf. The distance away from the shelf can be determined from LIDAR readings and/or identifying and triangulating

overlapping portions of image data from multiple cameras. Trigonometric functions can then be executed to calculate an actual, physical size of the empty space on the shelf using the determined distance value. For example, a function can include multiplying cosin of the angle from the shelf by the distance value, where the distance value can represent a multiplier for shelf size.

(118) As described above, a size of the buffer region (e.g., number of pixels) can vary depending on the distance of the camera from the shelf as well as the angle of the camera relative to the shelf. The greater the angle and the farther away, the larger the buffer region (e.g., the more pixels in the image data that will be ignored by the computer system when processing the image data). The smaller the angle and the closer to the shelf, the smaller the buffer region (e.g., the fewer pixels in the image data that will be ignored by the computer system).

(119) The computer system isolates the buffer region in image data captured by the camera in **1410**. Isolating the buffer region can include ignoring the buffer region when processing the image data. After all, the buffer region represents pixels in the image data that may not accurately depict part of a back of the shelf. Thus, by ignoring the buffer region, the computer system can reduce false positives that may arise if the computer system processes the image data without ignoring the buffer region.

(120) The computer system processes the image data surrounding the buffer region (**1412**). The computer system can, for example, process regions of the image data appearing above the buffer region. Such regions can more accurately approximate the back of the shelf in the image data. These regions can therefore be processed to determine whether sufficient backing of the shelf is visible to warrant an out of stock determination to be made for the particular shelf.

(121) FIGS. **15A-B** show example configurations of an apparatus **1502** on a cart **1500** to detect inventory conditions. FIG. **15A** depicts a side view of the apparatus **1502** on the cart **1500**. FIG. **15B** depicts a top down view of the apparatus **1502** on the cart **1500** as the cart **1500** moves through a physical environment, such as a retail store. It shall be noted that the apparatus **1502** is mobile and a user of any size and/or build can attach the apparatus **1502** to a cart, such as the cart **1500**, and remove the apparatus **1502** from the cart. The apparatus **1502** can, for example, be lightweight and may include one or more clamps to attach the apparatus **1502** to a side or sides of the cart **1500**. Additionally, the user can attach any quantity of the apparatus **1502** to the cart **1500**. For example, an apparatus **1502** can be placed at each corner of the cart **1500**. As shown in FIG. **15B**, an apparatus **1502** can be placed at each front corner of the cart **1500**. One or more apparatus **1502** can also be placed along sides of the cart **1500**. Moreover, in some implementations, the apparatus **1502** can be clamped or otherwise mounted to one or more tiers (e.g., baskets) of the cart **1500**, such as a bottom tier, a middle tier, and a top tier. As shown in FIG. **15A**, for example, the apparatus **1502** can be mounted to furthest corners of the bottom tier of the cart **1500** so as to reduce any potential awkwardness for users who are using the cart **1500** (e.g., loading products into the cart **1500**, removing products from the cart **1500**, etc.). As another example, the apparatus **1502** can be mounted to a far right or far left side of a handle at the back of the cart **1500** to make the cart **1500** easy to operate by the users.

(122) Referring to FIG. **15A**, the apparatus **1502** can be a vertical pole that attaches to a portion of the cart **1500**. For example, the apparatus **1502** can attach to one or more corners of the cart **1500**, such as at a front right and/or a front left corner of the cart **1500**. The apparatus **1502** can also be attached anywhere along right and/or left sides of the cart **1500** in some implementations. One or more of the components described herein in the system can be integrated into or otherwise attached to the apparatus **1502**.

(123) The apparatus **1502** may include an indoor location device **1504**, one or more lights **108A-N**, a high resolution camera **1510**, a low resolution camera **1512**, an output indicator **1514**, and a power source **1506**. One or more of **1504**, **108A-N**, **1510**, **1512**, **1514**, and **1506** can be optional. Moreover, the components **1504**, **108A-N**, **1510**, **1512**, **1514**, and **1506** can be arranged in any desired configuration.

(124) The indoor location device **1504** can be positioned at a top of the apparatus **1502** and pointed upwards towards a ceiling of the retail environment. As described above, the indoor location device **1504** can be an upward facing camera, configured to detect a location of the cart **1500** based on identifying lights in the ceiling of the retail environment. One or more other location positioning methods can be used to determine a current location of the cart **1500** in the retail environment.

(125) One or more of the lights **1508A-N** can be configured along a length of the apparatus **1502**. The lights **1508A-N** can be selectively actuated (e.g., by a cart controller, not depicted in FIGS. **15A-B**) to illuminate an area that is imaged by the cameras **1510** and/or **1512**. In the example apparatus **1502** of FIG. **15A**, there are two lights **1508A-N**. The light **1508A** is positioned from a top portion of the apparatus **1502** to a midpoint of the apparatus **1502**. The light **1508N** is positioned from slightly below the midpoint of the apparatus **1502** to a bottom portion of the apparatus **1502**. As a result, the lights **1508A-N** can illuminate an entire length/height of shelves or another region in the retail environment that may be imaged by the high resolution camera **1510** and/or the low resolution camera **1512**. One or more other arrangements of the lights **1508A-N** can be utilized. Moreover, the lights **1508A-N** can include infrared lights. In some implementations, the lights **1508A-N** can be white lights or other visible lights.

(126) The high resolution camera **1510** can be a PTZ camera or other movable camera as described throughout this disclosure. The low resolution camera **1512** can be a fixed camera as described throughout this disclosure. The output indicator **1514** can output a light or other signal that can be viewed by a user of the cart **1500**. The signal can indicate whether components of the apparatus **1502** are functioning properly. For example, if the high resolution camera **1510** is unable to adjust one or more movements (e.g., pan, tilt, zoom, focal length, aperture, shutter speed, etc.), then the camera **1510** can transmit a notification to the cart controller and/or the high resolution camera controller (not depicted in FIG. **15A**). The controller can instruct the output indicator **1514** to output a red light, which can indicate to the user that one or more components of the apparatus **1502** are not functioning properly. The user can then service or otherwise check the components. The output indicator **1514** can output a green light whenever components described herein are functioning properly. As mentioned, one or more other indicators can be outputted to depict operational status of components in the apparatus **1502** and/or the cart **1500**.

(127) Finally, the power source **1506** can be integrated into the apparatus **1502**. The power source **1506** can also be external to the apparatus **1502** and communicably coupled to one or more components of the apparatus **1502**. The power source **1506** can be replaceable, and/or rechargeable. For example, the power source **1506** can be a rechargeable battery that can be charged and/or replaced when it is lower on power and may no longer provide enough power to one or more of the components in the apparatus **1502**. In some implementations, the power source **1506** can plug into an outlet or other power source for charging purposes. When the power source **1506** is plugged in, the apparatus **1502** can be assumed to be outside of a main usage mode. The main usage mode can be when the cart **1500** is moving around an environment and images are being captured of the environment. During the main usage mode, only some image data may be transmitted to another computer system as described throughout this disclosure. For example, only image data demonstrating empty spots on shelves, price discrepancies, etc. can be transmitted to the computer system for further processing. Other image data, which can be used for training purposes or other algorithms implemented in the environment can then be transmitted to the computer system when the apparatus **1502** is not in the main usage mode. When not in the main usage mode (for example, the apparatus **1502** can be in batch mode), any and/or all data that is captured by components of the apparatus **1502** can be packaged and transmitted to the computer system described throughout this disclosure for further processing and analysis of out of stock conditions. Therefore, while the power source **1506** is charging, larger batches of data can be transmitted across networks.

(128) Refer to FIG. **9** for further discussion about the components of the apparatus **1502**.

(129) Referring to FIG. **15B**, the cart **1500** can be moving in an environment **1522**. The

environment **1522** can be a retail environment, such as a store, as described previously. The cart **1500** can be pushed by a user, such as a store employee. The cart **1500** can also be autonomously operated (e.g., by a robot) and can automatically move throughout the environment **1522**. The cart **1500** can move in the environment **1522** and pass shelves **1518A** and **1518B**. The shelves **1518A** and **1518B** can include products, that can be in stock or out of stock, as described herein.

(130) In the example of FIG. **15B**, the cart **1500** includes two apparatuses **1502A** and **1502B**. Each of the apparatuses **1502A** and **1502B** are vertical poles that are attached to front corners of the cart **1500**. The apparatuses **1502A** and **1502B** are described further in FIG. **15A**. As the cart **1500** moves in direction **1516**, cameras along each of the apparatuses **1502A** and **1502B** can capture image data of the respective shelves **1518A** and **1518B** in their respective fields of view **1520A** and **1520B**.

(131) As an illustrative example, a high resolution camera on the apparatus **1502A** can be configured to pan left and right (or perform other operations) to capture image data of the shelf **1518A** in the camera's field of view **1520A** while the cart **1500** is moving past the shelf **1518A**. Similarly, a high resolution camera on the apparatus **1502B** can be configured to tilt and zoom (or perform other operations) to capture image data of the shelf **1518B** in the camera's field of view **1520B** while the cart **1500** is moving past the shelf **1518B**.

(132) In some implementations, the cameras on the apparatuses **1502A** and **1502B** can be operated simultaneously by the cart controller and/or one or more high resolution camera controllers. In some implementations, only components of one of the apparatuses **1502A** and **1502B** may be selectively controlled and/or operated. In yet some implementations, components of the apparatus **1502A** can be selectively controlled while components of the apparatus **1502B** are in a sleep mode or other mode in which they are not performing operations described throughout this disclosure.

(133) In some implementations, where the cart **1500** only has one apparatus **1502A**, the cart **1500** may move down the aisle such that the camera on the apparatus **1502A** captures image data of the shelf **1518A** in the field of view **1520A**. The cart **1500** can then be turned around and may move in a direction opposite the direction **1516** such that the camera on the apparatus **1502A** can capture image data of the shelf **1518B** in the field of view **1520A**.

(134) FIG. **16** is a flowchart of an example process **1600** for determining whether to use image data or RFID sensor data to detect inventory conditions. In the process **1600**, a cart such as the cart **106** or the cart **1500** can use a system **900** to collect image data and/or RFID sensor data that can be used for detecting inventory conditions. However, other devices and systems can be used to perform the process **1600** and other processes described throughout this disclosure.

(135) Referring to the process **1600**, location information is received **1602**. For example, the cart controller depicted and described in reference to FIG. **9** can receive location information from one or more components, such as the indoor location device, the inertia measurement unit, one or more of the low resolution cameras, a depth sensor, and/or one or more other devices or sensors that may be positioned throughout an environment where the cart (e.g., mobile apparatus) is located. In some implementations, the cart controller can receive sensor data that can be utilized by the cart controller to determine location information (e.g., a current location) of the cart.

(136) Next, the cart controller determines a region in the store where the cart is located based on the location information (**1604**). The cart controller can, for example, compare the location information to information in a lookup table. The lookup table can correlate locations (e.g., GPS coordinates, other location-based signals) to actual regions in the environment, such as departments (e.g., groceries, electronics, clothes, furniture, etc.), particular aisles, etc.

(137) The cart controller determines whether the region uses images to determine stock status in **1606**. The lookup table can also associate each region with type of products/items and preference for image data or RFID sensor data. Databases or other structured files can also contain associations between regions in the environment and type of data that can be used for determining stock status.

(138) When the products have definite physical structures, image data can be preferred since physical shapes of the products can be more accurately extracted from the image data to determine whether a particular shelf has an out of stock condition. On the other hand, products that do not have definite physical structures may be more challenging to analyze for stock status using image data. As a result, RFID sensor data can be preferred for use in determining a stock status of those products.

(139) As an illustrative example, boxes of cereal can be easily detected from image data for having definite, rigid structures. Those definite, rigid structures can be more easily identified and extracted from an image that contains the cereal boxes on a shelf. On the other hand, clothes can be folded and stacked on a shelf, but sometimes, the stacks of clothes may not have a uniform or definite shape or structure. Clothes can be stacked or heaped on top of each other and may form less definite shapes such as piles of fabric. As a result, identifying the clothes in image data can be more challenging. Instead, RFID sensor data can be used to determine how much of items such as clothes are located on their respective shelf and whether there is an out of stock condition.

(140) If the region uses images for determining stock status, then the cart controller instructs one or more of the cameras to capture images of the shelves in the region (**1608**). In other words, the region contains products that typically have definitive physical structures. The region can be a cereal and breakfast aisle in a grocery store, by way of example.

(141) Once the camera(s) captures images of the shelf, the camera(s) transmits the image data to the cart controller. The cart controller processes the received data as described throughout this disclosure to determine stock status (**1612**).

(142) If the region uses RFID sensor data for determining stock status, then the cart controller instructs one or more RFID sensor devices to detect products in the region (**1610**). In other words, the region contains products that typically have non-definitive physical structures. The region can be a clothing department in a retail store, by way of example.

(143) In some implementations, the RFID sensor devices may be continuously capturing the RFID sensor data regardless of performance of **1602-1606** described above. Therefore, in **1610**, the cart controller can simply determine that the RFID sensor data should be used to determine stock status and thus request the RFID sensor data from the RFID sensor devices for use in the determination. In some implementations, the RFID sensor devices may only be turned on and instructed to capture the RFID sensor data when the cart is in a region that does not use images to determine stock status (**1606**).

(144) Moreover, in some implementations, the cameras may not be continuously capturing images. Rather, the cart controller can selectively turn on the cameras in **1608** when the cart controller determines that the region is one in which image data should be used for determining stock status. Only activating the cameras when they are needed for stock status determinations can save processing power and compute resources. Once the cart controller determines that the cart is located in a region of the store that does not need image data for stock status determinations, the cart controller can turn off the cameras or otherwise instruct the cameras to stop capturing images. Similarly, if the cart controller determines that the cart is standing idle (e.g., the cart has not moved for at least a predetermined period of time), the cart controller can instruct the cameras to stop capturing images to save on power consumption, processing power, and compute resources.

(145) In some implementations, the cameras can be configured to continuously capture images as the cart moves throughout the store, however the images may not all be used for stock status determinations. The images can be continuously captured and if the cart controller determines in **1606** that the region uses images to determine stock status, then the cart controller can request the images of the region from the cameras and use just those images for this determination. Once the cart is standing idle or is plugged in and charging, as described in reference to FIGS. 15A-B, the cart controller can request, from the cameras, all of the images that were captured while the cart was moving. All of the images can then be used by the cart controller or another computer system

for other processing and/or analysis of conditions in the store.

(146) Once the RFID sensor(s) detect product data, the sensor(s) transmits the sensor data to the cart controller. The cart controller processes the received data as described throughout this disclosure to determine stock status (**1612**).

(147) FIG. **17** shows an example of a computing device **1700** and an example of a mobile computing device that can be used to implement the techniques described here. The computing device **1700** is intended to represent various forms of digital computers, such as laptops, desktops, workstations, personal digital assistants, servers, blade servers, mainframes, and other appropriate computers. The mobile computing device is intended to represent various forms of mobile devices, such as personal digital assistants, cellular telephones, smart-phones, and other similar computing devices. The components shown here, their connections and relationships, and their functions, are meant to be exemplary only, and are not meant to limit implementations of the inventions described and/or claimed in this document.

(148) The computing device **1700** includes a processor **1702**, a memory **1704**, a storage device **1706**, a high-speed interface **1708** connecting to the memory **1704** and multiple high-speed expansion ports **1710**, and a low-speed interface **1712** connecting to a low-speed expansion port **1714** and the storage device **1706**. Each of the processor **1702**, the memory **1704**, the storage device **1706**, the high-speed interface **1708**, the high-speed expansion ports **1710**, and the low-speed interface **1712**, are interconnected using various busses, and can be mounted on a common motherboard or in other manners as appropriate. The processor **1702** can process instructions for execution within the computing device **1700**, including instructions stored in the memory **1704** or on the storage device **1706** to display graphical information for a GUI on an external input/output device, such as a display **1716** coupled to the high-speed interface **1708**. In other implementations, multiple processors and/or multiple buses can be used, as appropriate, along with multiple memories and types of memory. Also, multiple computing devices can be connected, with each device providing portions of the necessary operations (e.g., as a server bank, a group of blade servers, or a multi-processor system).

(149) The memory **1704** stores information within the computing device **1700**. In some implementations, the memory **1704** is a volatile memory unit or units. In some implementations, the memory **1704** is a non-volatile memory unit or units. The memory **1704** can also be another form of computer-readable medium, such as a magnetic or optical disk.

(150) The storage device **1706** is capable of providing mass storage for the computing device **1700**. In some implementations, the storage device **1706** can be or contain a computer-readable medium, such as a floppy disk device, a hard disk device, an optical disk device, or a tape device, a flash memory or other similar solid state memory device, or an array of devices, including devices in a storage area network or other configurations. A computer program product can be tangibly embodied in an information carrier. The computer program product can also contain instructions that, when executed, perform one or more methods, such as those described above. The computer program product can also be tangibly embodied in a computer- or machine-readable medium, such as the memory **1704**, the storage device **1706**, or memory on the processor **1702**.

(151) The high-speed interface **1708** manages bandwidth-intensive operations for the computing device **1700**, while the low-speed interface **1712** manages lower bandwidth-intensive operations. Such allocation of functions is exemplary only. In some implementations, the high-speed interface **1708** is coupled to the memory **1704**, the display **1716** (e.g., through a graphics processor or accelerator), and to the high-speed expansion ports **1710**, which can accept various expansion cards (not shown). In the implementation, the low-speed interface **1712** is coupled to the storage device **1706** and the low-speed expansion port **1714**. The low-speed expansion port **1714**, which can include various communication ports (e.g., USB, Bluetooth, Ethernet, wireless Ethernet) can be coupled to one or more input/output devices, such as a keyboard, a pointing device, a scanner, or a networking device such as a switch or router, e.g., through a network adapter.

(152) The computing device **1700** can be implemented in a number of different forms, as shown in the figure. For example, it can be implemented as a standard server **1720**, or multiple times in a group of such servers. In addition, it can be implemented in a personal computer such as a laptop computer **1722**. It can also be implemented as part of a rack server system **1724**. Alternatively, components from the computing device **1700** can be combined with other components in a mobile device (not shown), such as a mobile computing device **1750**. Each of such devices can contain one or more of the computing device **1700** and the mobile computing device **1750**, and an entire system can be made up of multiple computing devices communicating with each other.

(153) The mobile computing device **1750** includes a processor **1752**, a memory **1764**, an input/output device such as a display **1754**, a communication interface **1766**, and a transceiver **1768**, among other components. The mobile computing device **1750** can also be provided with a storage device, such as a micro-drive or other device, to provide additional storage. Each of the processor **1752**, the memory **1764**, the display **1754**, the communication interface **1766**, and the transceiver **1768**, are interconnected using various buses, and several of the components can be mounted on a common motherboard or in other manners as appropriate.

(154) The processor **1752** can execute instructions within the mobile computing device **1750**, including instructions stored in the memory **1764**. The processor **1752** can be implemented as a chipset of chips that include separate and multiple analog and digital processors. The processor **1752** can provide, for example, for coordination of the other components of the mobile computing device **1750**, such as control of user interfaces, applications run by the mobile computing device **1750**, and wireless communication by the mobile computing device **1750**.

(155) The processor **1752** can communicate with a user through a control interface **1758** and a display interface **1756** coupled to the display **1754**. The display **1754** can be, for example, a TFT (Thin-Film-Transistor Liquid Crystal Display) display or an OLED (Organic Light Emitting Diode) display, or other appropriate display technology. The display interface **1756** can comprise appropriate circuitry for driving the display **1754** to present graphical and other information to a user. The control interface **1758** can receive commands from a user and convert them for submission to the processor **1752**. In addition, an external interface **1762** can provide communication with the processor **1752**, so as to enable near area communication of the mobile computing device **1750** with other devices. The external interface **1762** can provide, for example, for wired communication in some implementations, or for wireless communication in other implementations, and multiple interfaces can also be used.

(156) The memory **1764** stores information within the mobile computing device **1750**. The memory **1764** can be implemented as one or more of a computer-readable medium or media, a volatile memory unit or units, or a non-volatile memory unit or units. An expansion memory **1774** can also be provided and connected to the mobile computing device **1750** through an expansion interface **1772**, which can include, for example, a SIMM (Single In Line Memory Module) card interface. The expansion memory **1774** can provide extra storage space for the mobile computing device **1750**, or can also store applications or other information for the mobile computing device **1750**. Specifically, the expansion memory **1774** can include instructions to carry out or supplement the processes described above, and can include secure information also. Thus, for example, the expansion memory **1774** can be provide as a security module for the mobile computing device **1750**, and can be programmed with instructions that permit secure use of the mobile computing device **1750**. In addition, secure applications can be provided via the SIMM cards, along with additional information, such as placing identifying information on the SIMM card in a non-hackable manner.

(157) The memory can include, for example, flash memory and/or NVRAM memory (non-volatile random access memory), as discussed below. In some implementations, a computer program product is tangibly embodied in an information carrier. The computer program product contains instructions that, when executed, perform one or more methods, such as those described above. The

computer program product can be a computer- or machine-readable medium, such as the memory **1764**, the expansion memory **1774**, or memory on the processor **1752**. In some implementations, the computer program product can be received in a propagated signal, for example, over the transceiver **1768** or the external interface **1762**.

(158) The mobile computing device **1750** can communicate wirelessly through the communication interface **1766**, which can include digital signal processing circuitry where necessary. The communication interface **1766** can provide for communications under various modes or protocols, such as GSM voice calls (Global System for Mobile communications), SMS (Short Message Service), EMS (Enhanced Messaging Service), or MMS messaging (Multimedia Messaging Service), CDMA (code division multiple access), TDMA (time division multiple access), PDC (Personal Digital Cellular), WCDMA (Wideband Code Division Multiple Access), CDMA2000, or GPRS (General Packet Radio Service), among others. Such communication can occur, for example, through the transceiver **1768** using a radio-frequency. In addition, short-range communication can occur, such as using a Bluetooth, WiFi, or other such transceiver (not shown). In addition, a GPS (Global Positioning System) receiver module **1770** can provide additional navigation- and location-related wireless data to the mobile computing device **1750**, which can be used as appropriate by applications running on the mobile computing device **1750**.

(159) The mobile computing device **1750** can also communicate audibly using an audio codec **1760**, which can receive spoken information from a user and convert it to usable digital information. The audio codec **1760** can likewise generate audible sound for a user, such as through a speaker, e.g., in a handset of the mobile computing device **1750**. Such sound can include sound from voice telephone calls, can include recorded sound (e.g., voice messages, music files, etc.) and can also include sound generated by applications operating on the mobile computing device **1750**.

(160) The mobile computing device **1750** can be implemented in a number of different forms, as shown in the figure. For example, it can be implemented as a cellular telephone **1780**. It can also be implemented as part of a smart-phone **1782**, personal digital assistant, or other similar mobile device.

(161) Various implementations of the systems and techniques described here can be realized in digital electronic circuitry, integrated circuitry, specially designed ASICs (application specific integrated circuits), computer hardware, firmware, software, and/or combinations thereof. These various implementations can include implementation in one or more computer programs that are executable and/or interpretable on a programmable system including at least one programmable processor, which can be special or general purpose, coupled to receive data and instructions from, and to transmit data and instructions to, a storage system, at least one input device, and at least one output device.

(162) These computer programs (also known as programs, software, software applications or code) include machine instructions for a programmable processor, and can be implemented in a high-level procedural and/or object-oriented programming language, and/or in assembly/machine language. As used herein, the terms machine-readable medium and computer-readable medium refer to any computer program product, apparatus and/or device (e.g., magnetic discs, optical disks, memory, Programmable Logic Devices (PLDs)) used to provide machine instructions and/or data to a programmable processor, including a machine-readable medium that receives machine instructions as a machine-readable signal. The term machine-readable signal refers to any signal used to provide machine instructions and/or data to a programmable processor.

(163) To provide for interaction with a user, the systems and techniques described here can be implemented on a computer having a display device (e.g., a CRT (cathode ray tube) or LCD (liquid crystal display) monitor) for displaying information to the user and a keyboard and a pointing device (e.g., a mouse or a trackball) by which the user can provide input to the computer. Other kinds of devices can be used to provide for interaction with a user as well; for example, feedback provided to the user can be any form of sensory feedback (e.g., visual feedback, auditory feedback,

or tactile feedback); and input from the user can be received in any form, including acoustic, speech, or tactile input.

(164) The systems and techniques described here can be implemented in a computing system that includes a back end component (e.g., as a data server), or that includes a middleware component (e.g., an application server), or that includes a front end component (e.g., a client computer having a graphical user interface or a Web browser through which a user can interact with an implementation of the systems and techniques described here), or any combination of such back end, middleware, or front end components. The components of the system can be interconnected by any form or medium of digital data communication (e.g., a communication network). Examples of communication networks include a local area network (LAN), a wide area network (WAN), and the Internet.

(165) The computing system can include clients and servers. A client and server are generally remote from each other and typically interact through a communication network. The relationship of client and server arises by virtue of computer programs running on the respective computers and having a client-server relationship to each other.

Claims

1. An apparatus for determining an out of stock condition on a shelf, the apparatus comprising: a camera in data communication with one or more processors; the one or more processors; a mobile power source providing power to the camera and to the one or more processors; a network interface for sending and receiving messages with physically remote destinations over a data network; and computer memory storing instructions that, when executed by the one or more processors, cause the one or more processors to perform operations comprising: receiving image data from the camera comprising a plurality of pixel locations, each pixel location being uniquely addressed in the image data and storing a pixel value captured by the camera at that location; generating a backing map comprising a plurality of cell locations, each cell location being uniquely addressed in the backing map and sharing a unique address with a corresponding pixel location in the image data, each cell location storing a backing value being an empty value if the pixel value is classified as showing the backing of a shelf and the backing value being a nonempty value if the pixel value is classified as not showing the backing of the shelf; determining, in the backing map, a shelf area representing a location of a shelf captured by the camera; identifying, in the backing map, an empty area by finding an area above the shelf area containing a threshold number of cell locations storing the empty value; determining a shelf-identifier for a shelf that is empty of items based on the empty area in the backing map; and generating, using the shelf-identifier, instructions to restock the shelf that is empty of items.

2. The apparatus of claim 1, wherein the operations further comprise: identifying a barcode target area in the shelf area based on the location of the empty area; and reading, in the image data, a barcode located in the image data based on the location of the barcode target area in the shelf area.

3. The apparatus of claim 1, wherein the operations further comprise: transmitting to an inventory server system through the network interface an empty-shelf message that includes the shelf-identifier.

4. The apparatus of claim 1, wherein generating the backing map comprises: for each cell location: accessing the pixel value of the corresponding pixel location with the cell location's unique address; generating a backing value by providing, to a classifier, the pixel value of the corresponding pixel location; receiving, from the classifier, the backing value; and recording, in the cell location, the backing value.

5. The apparatus of claim 4, wherein generating a backing value by providing, to a classifier, the pixel value of the corresponding pixel location further comprises providing the image data and the unique address of the cell location; wherein the classifier is configured to generate the backing

value using a model that receives, as input, at least i) the pixel value of the corresponding pixel location and ii) other pixel values in the image data other than the pixel value of the corresponding pixel location based on the unique address.

6. The apparatus of claim 4, wherein generating a backing value by providing, to a classifier, the pixel value of the corresponding pixel location further comprises providing a facility location that defines i) where in a facility the image data was captured and ii) item location data that specifies shelves in the facility and objects to be stored on the shelves in the facility; and wherein the classifier is configured to generate the backing value using a model that receives, as input, at least i) the pixel value of the corresponding pixel location and ii) objects to be stored near the facility location that defines where in a facility the image data was captured.

7. The apparatus of claim 4, wherein the classifier is a machine-learning classifier.

8. The apparatus of claim 7, wherein the apparatus is in data communication with a server system configured to: receive records of false-empty results; generate supplemental training data from the records of false-empty results; and retrain the classifier using the supplemental training data.

9. The apparatus of claim 8, wherein to retrain the classifier using the supplemental training data, the server system is configured to retrain the classifier using at least some original training data that was used to train the classifier in generating the false-empty results.

10. The apparatus of claim 1, further comprising: a fixed camera being fixedly mounted on the cart at a first angle; wherein the camera is a high resolution camera being controllably mounted on the cart, the high resolution camera configured to capture high resolution images and to engage at least one of pan, tilt, and zoom operations in response to engagement instructions received from a high resolution camera controller; and a high resolution camera controller configured to: receive high resolution camera instructions; and responsive to receiving the high resolution camera instructions, send the engagement instructions to the high resolution camera; wherein the operations further comprise: receiving, from the fixed camera, first image data that captures a first inventory object; determining, from the first image data, a spatial location of a first inventory object; generating high resolution camera instructions that are configured to cause the high resolution camera to capture the first inventory object; transmitting the high resolution camera instructions to the high resolution camera controller; and receiving, from the high resolution camera, the image data.

11. The apparatus of claim 10, wherein: the high resolution camera is configured to operate in a plurality of modes, the plurality of modes comprising at least i) a sleep mode and ii) a working mode, wherein the working mode consumes more power than the sleep mode, and the high resolution camera controller is configured to transition, in response to receiving the high resolution camera instructions, the high resolution camera from the sleep mode to the working mode and to later transition the high resolution camera from the working mode to the sleep mode after the image data is received from the high resolution camera.

12. A method for determining an out of stock condition on a shelf, the method comprising: receiving, by a computing system, image data from a camera comprising a plurality of pixel locations, each pixel location being uniquely addressed in the image data and storing a pixel value captured by a camera at that location; generating, by the computing system, a backing map comprising a plurality of cell locations, each cell location being uniquely addressed in the backing map and sharing a unique address with a corresponding pixel location in the image data, each cell location storing a backing value being an empty value if the pixel value is classified as showing the backing of a shelf and the backing value being a nonempty value if the pixel value is classified as not showing the backing of the shelf; determining, by the computing system and in the backing map, a shelf area representing a location of the shelf captured by the camera; identifying, by the computing system and in the backing map, an empty area by finding an area above the shelf area containing a threshold number of cell locations storing the empty value; determining, by the computing system, a shelf-identifier for a shelf that is empty of items based on the empty area in the backing map; and generating, by the computing system and using the shelf-identifier,

instructions to restock the shelf that is empty of items.

13. The method of claim 12, the method further comprising: identifying, by the computing system, a barcode target area in the shelf area based on the location of the empty area; and reading, by the computing system and in the image data, a barcode located in the image data based on the location of the barcode target area in the shelf area.

14. The method of claim 12, the method further comprising: transmitting, by the computing system and to an inventory server system, an empty-shelf message that includes the shelf-identifier.

15. The method of claim 12, wherein generating, by the computing system, the backing map comprises: for each cell location: accessing the pixel value of the corresponding pixel location with the cell location's unique address; generating a backing value by providing, to a classifier, the pixel value of the corresponding pixel location; receiving, from the classifier, the backing value; and recording, in the cell location, the backing value.

16. The method of claim 15, wherein the classifier is a machine-learning classifier.

17. A system for determining an out of stock condition on a shelf, the system comprising: one or more processors; and computer memory storing instructions that, when executed by the one or more processors, cause the one or more processors to perform operations comprising: receiving image data comprising a plurality of pixel locations, each pixel location being uniquely addressed in the image data and storing a pixel value captured by a camera at that location; generating a backing map comprising a plurality of cell locations, each cell location being uniquely addressed in the backing map and sharing a unique address with a corresponding pixel location in the image data, each cell location storing a backing value being an empty value if the pixel value is classified as showing the backing of a shelf and the backing value being a nonempty value if the pixel value is classified as not showing the backing of the shelf; determining, in the backing map, a shelf area representing a location of the shelf captured by the camera; and identifying, in the backing map, an empty area by finding an area above the shelf area containing a threshold number of cell locations storing the empty value; determining a shelf-identifier for a shelf that is empty of items based on the empty area in the backing map; and generating, using the shelf-identifier, instructions to restock the shelf that is empty of items.

18. The system of claim 17, the operations further comprising: identifying a barcode target area in the shelf area based on the location of the empty area; and reading, in the image data, a barcode located in the image data based on the location of the barcode target area in the shelf area.

19. The system of claim 17, the operations further comprising: transmitting, to an inventory server system, an empty-shelf message that includes the shelf-identifier.

20. The system of claim 17, wherein generating the backing map comprises: for each cell location: accessing the pixel value of the corresponding pixel location with the cell location's unique address; generating a backing value by providing, to a classifier, the pixel value of the corresponding pixel location; receiving, from the classifier, the backing value; and recording, in the cell location, the backing value.
